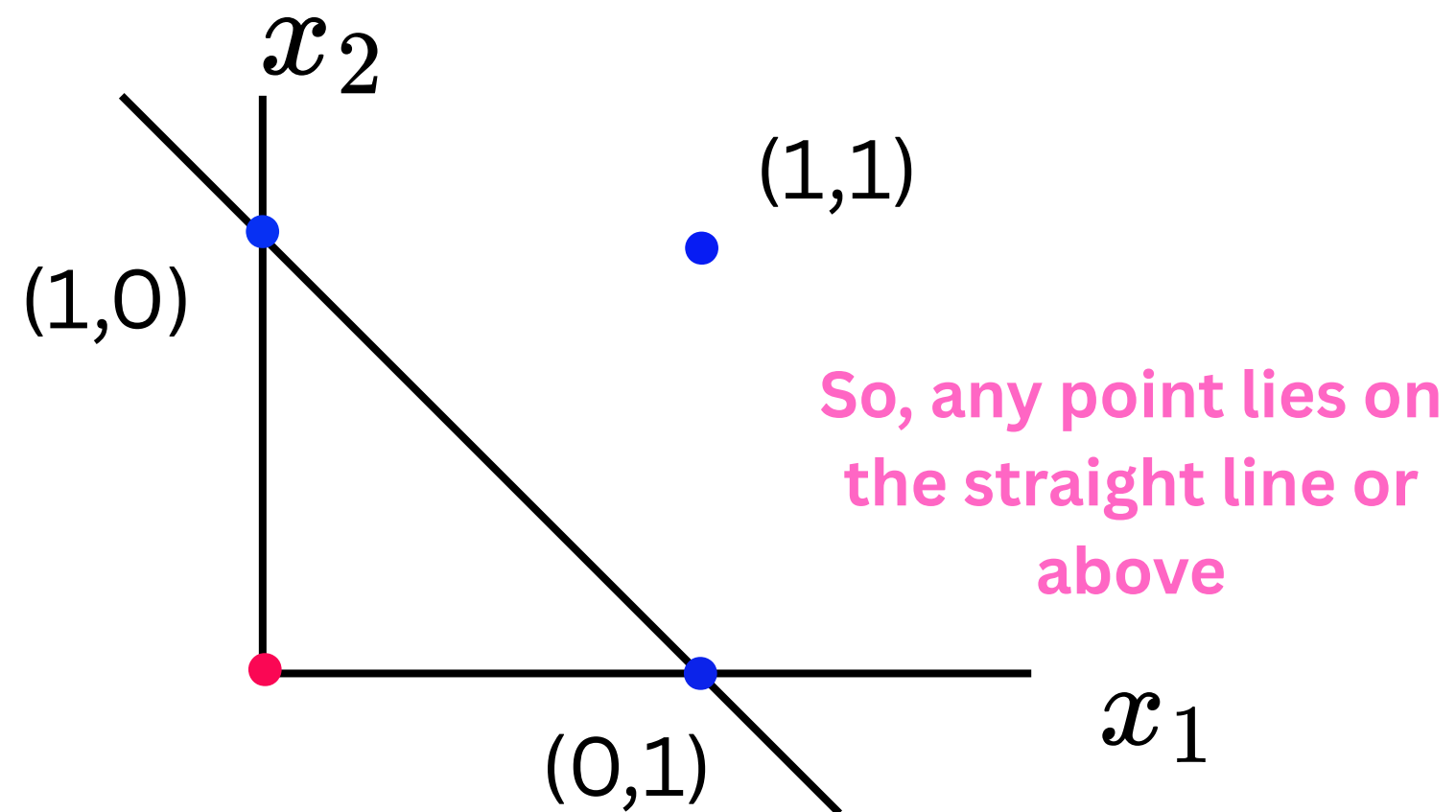




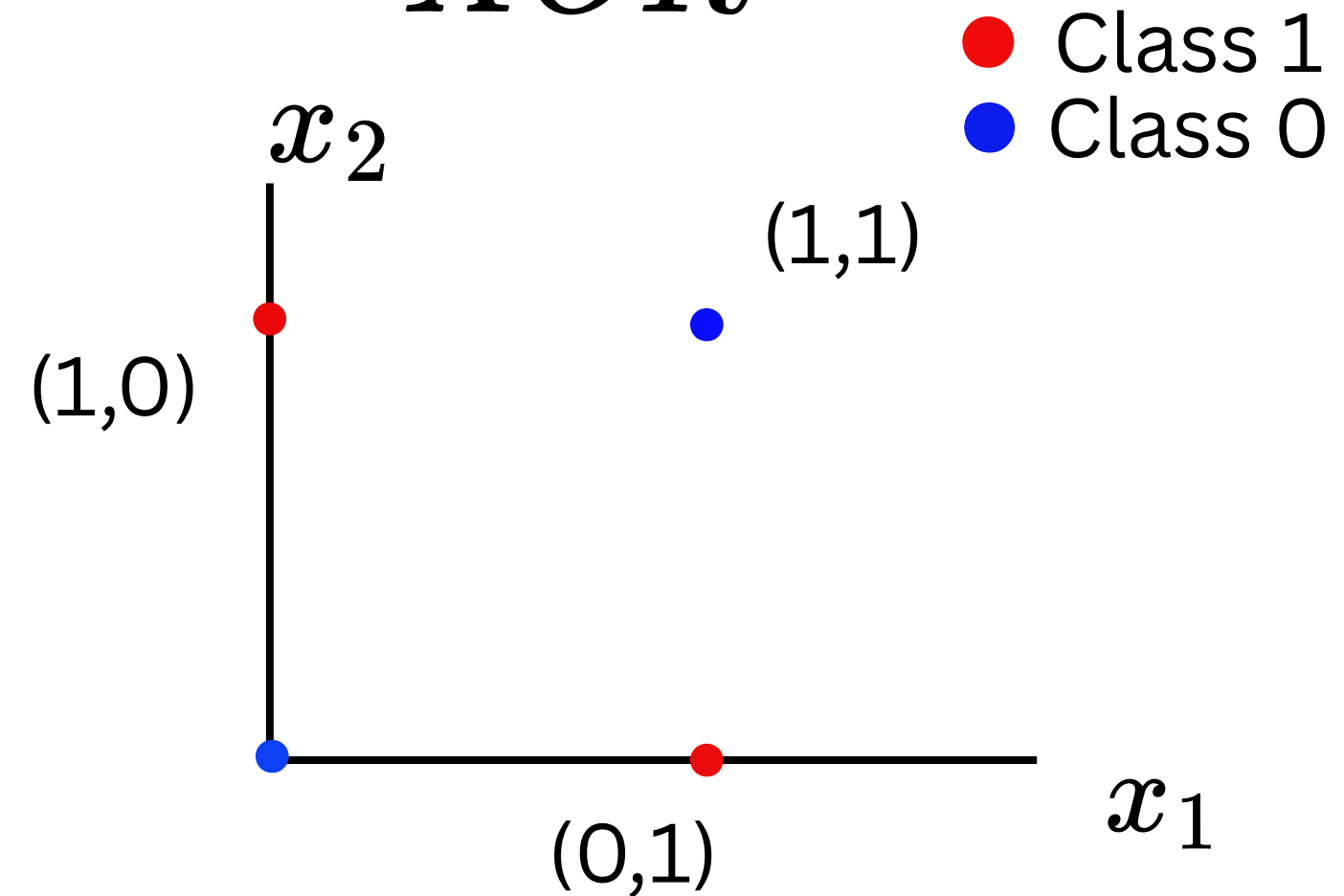
Separability and Non-Separable problem

OR $\theta \geq 1$



- In above scenario we can easily draw a line to separate these two classes

XOR



- In above scenario there is no way that we can draw a line to separate these two classes
- The problem is called XOR problem



Multi-Layer Perceptron

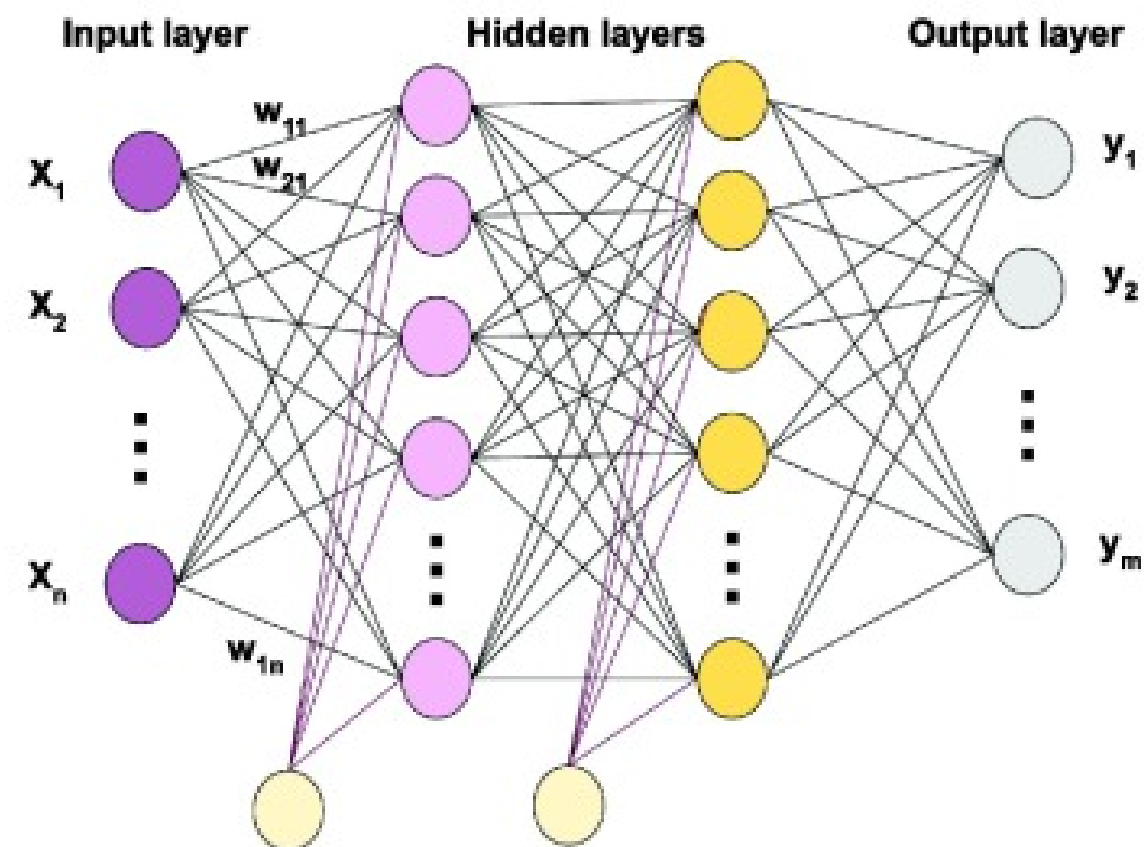
A **Multi-Layer Perceptron (MLP)** is a feed-forward neural network made of:

- Input layer
- One or more hidden layers
- Output layer
- Activation functions
- It is the simplest form of a deep neural network.



Multi-Layer Perceptron

Structure of MLP



(a) Input Layer

- Receives features

Example: for an image patch with 64 pixels $\rightarrow$ 64 input neurons

(b) Hidden Layers

Each neuron performs:

$$z = w^T x + b, \quad a = f(z)$$

Where

- w = weights
- b = bias
- f = activation function (ReLU, tanh, sigmoid, etc.)

(c) Output Layer

- Sigmoid $\rightarrow$ binary classification
- Softmax $\rightarrow$ multi-class
- Linear $\rightarrow$ regression



Multi-Layer Perceptron

Forward Pass

Information flows left → right.

For layer l :

$$z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}$$

$$a^{(l)} = f(z^{(l)})$$

Backward Pass

MLP learns weights by minimizing loss using gradient descent:

$$W \leftarrow W - \eta \frac{\partial L}{\partial W}$$

Steps:

1. Forward pass
2. Compute loss
3. Backward pass (chain rule)
4. Update weights
5. Repeat → convergence

How Gradient Flow Works

Backpropagation in Neural Networks

